Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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(The names and affiliations SHOULD NOT be included in the draft submitted for review)
(leave blank up to line 10 - remove line numbering from final version)

Abstract

This document explains how to prepare a paper for submission to the Building Simulation and Optimisation Conference 2022. It also includes the instructions for submission and some other information.

Key Innovations

- Key innovations section added
- Practical implications section added

Practical Implications

This updated template allows practitioners to quickly grasp the practical implications of any paper. Use it and share your work to a broader audience!

Introduction

The paper should be prepared in A4 size, portrait oriented, with 25 mm margins at top and bottom, and 20 mm margins at left and right. Use font type Times (Times New Roman) for the entire document with different style at different part of the paper, as explained later. The **maximum number of pages is 8**. Please **do not** add page numbers, headers or footers.
The paper includes 2 sections (top section and main body) and different parts. Mandatory - in bold - and optional (recommended but not limited to) parts are:
Top section

- **Title**
- **Authors**
- **Authors' affiliations**

Main body

- **Abstract**
- **Key innovations**
- **Practical implications**
- **Introduction**
- Methods
- Results
- Discussion

- **Conclusion**
- Acknowledgments
- Nomenclature
- **References**

Optional parts can be replaced according to the needs but the order should be preserved.
Second level paragraphs are acceptable. Lower level paragraphs are discouraged.
Do not leave empty lines between titles, paragraphs and figures in the main body. Use paragraph spacing above and below titles and paragraphs as specified in the following.
In case you use MS-Word, you can refer to the predefined styles in the gallery included in the MS-word template (in alphabetical order):

- Authors
- Captions
- Equation
- Figure
- List (bulleted)
- List (numbered)
- Heading 1
- Heading 2
- Normal
- Reference
- Tables
- Title

Parts of the paper: top section

The top section of the first page of the paper is used for the title of the paper, list of authors, and authors' affiliation. This section always consists of 10 lines of exactly 14 points spacing, in single column format.

Title, authors, and authors' affiliations

Titles (MS-word: select *Title style*) should be in bold font size 12 points. Do not use more than 2 lines for the title, and try to limit it to ten words.
Authors (MS-word: select *Authors style*), authors' affiliations, and other information should be in font size 12 points. In case of more than one affiliation, reference superscripts after the author's names and before the corresponding affiliation will be added.
Authors' affiliations (MS-word: *always in Authors style*), can include contact (e-mails, telephone numbers, postal address, etc.) and other information. However, the 10 lines limitation applies. If you need less than 10 lines for all the information, please leave the remaining lines blank.



Gambar 1: A gentle reminder.

Main body

After the 10 lines in the top section, the rest of the document is in two-column format, with a space between the two columns of 80 mm. The space between the columns should be centred on the paper. Line spacing is exactly 12 points.
The main text should be typed in 10 point font, justified, with 3 points spacing below each paragraph (*MS-word: select Normal style*).
The headings of each section should use font size of 12 point, capitalized, with 3 and 3 points spacing above and below (*MS-word: select Heading 1 style*). The headings of the subsection should use 10 points size and bold, with 3 points spacing below (*MS-word: select Heading 2 style*). Left alignment is used for headings. Both the sections and subsections **should not** be numbered.

Abstract

The abstract of your paper should be about 100 words.
Abstracts must clearly identify (i) what is the current state of the art, (ii) what are its deficiencies, (iii) what methods have been applied, (iv) what are the results, and (v) what is the lasting contribution of the submission.

Key inovations

In 1-5 bullet points, clearly highlight where this paper moves beyond the state of the art.

Practical implications

The practical implications section should be about 20 - 50 words.
Capture succinctly what a simulation practitioner should take away from your paper: always do . . . ; avoid . . . ; consider x and beware of y in case of . . .

Figures and Tables

Figures and Tables are preferably included in the text where they are discussed rather than at the end of the paper. In the case of figures, select the “In line with text" from the layout options if you use MS-Word. A centred alignment with 3 and 3 points spacing above and below is required (*MS-word: select Figure style*).
Both figures and tables must have a number and caption. Figure 1 (*refer this way to a figure in the text, do not use Fig. 1 or figure 1*) is an example of a graph in the text. Captions (*MS-word: use Captions style*) always follow the figure in italic font size 10 points, and centred, with 3 and 3 points spacing above and below.
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Colour images are welcome.
Table 1 shows an example of a table, where the caption should be on the top of the table, with the same format as figure captions. Text in tables can be in font size of 9 points and centred, with

no spacing above and below (MS-word: use *Tables style*). Oversized figures and tables may be

Tabel 1: Example of a table.

Heading 1	**Heading 2**	**Heading 3**
Entry 1	Entry 2	Entry 3

included in the text. However, authors may arrange the layout properly so that it appears at the top or the bottom of the page, or preferably on a separate page at the end of the paper. Figure 2 is an example of an oversized figure.

Equations and units

Each significant equation or formula should be displayed on a separate line. Center equations and place consecutive equation numbers flush right in parentheses. For example,

$$a^2 + b^2 = c^2. \tag{1}$$

If you use MS-Word, use the Equation style and use tabs to position equations and equation numbers correctly. Mathematical symbols should be clear and avoid ambiguities. Keep using font size 10 points. Symbols representing physical quantities (or variables) are italic. Symbols representing units, operators, labels and numbers, are regular (upright).
Equations should be referenced in the text by their number (1). A brief description of the symbols used in your paper should be added in a nomenclature section at the end of the text or inline with the text where the symbols are first used.
SI units of measurement are mandatory. If other units are mentioned, always provide their equivalent value according to SI.
SI notation must be adopted as regards symbols, prefixes and other writing rules. In particular comma as a decimal separator is preferred to the point (which is anyhow allowed if used consistently throughout the manuscript). If needed, only blank spaces are allowed as thousand separators.

References

All publications cited in the text should be listed at the end of the Main body in a References section, in alphabetical order of the family name of the first author. From the second line on, each entry should be indented by 5 mm. Justified alignment with 3 and 3 points spacing above and below each paragraph must be used (*MS-word: select Reference style*). Examples of reference style are reported in the reference section below for journals, conference proceedings, books and technical standards.
In the main text, refer to a reference using author-year style such as Clarke2015:1 or ChuMajumdar2012. When the reference is not a part of the sentence, use the author-year style in brackets ([?, ?, ?, ?, ?, ?]).

Bulleted and numbered lists

Numbered and bulleted lists must be justified, with no indentation, and 5 mm hanging. You may use List (bulleted) or List (Numbered) styles in MS-Word.

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If you use LaTeX

Please use the LaTeXtemplate you can download from the conference website. In the compressed folder, you will find:

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- **BSO2022.bst**: this is the bibliography style of the BS2021 conference proceedings, which we are using for BSO 2022.

Submission instruction

1. To enable the blind review process please do not include your name and affiliation on the draft submitted for review. This information will be requested in the final submission.
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3. Papers must be at most 8 pages long.
4. You can use any file name, but please do not use special characters, umlaute, etc.
5. The maximum file size is 10 MB, but please use vector graphics where possible to reduce your file size.
6. If your paper has not met all the requirements for submission, your file **will not be processed for review and you will be requested to resubmit**. If everything is in order, the paper will be up-loaded to the review area of the conference web site.

Conclusion

This paper has shown how to prepare a paper for submission to Building Simulation and Optimisation Conference 2022. Good luck with your paper.

Acknowledgment

This document is a summary of various documents from previous Building Simulation Conferences.

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Building Simulation 2021

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